

PROJECT INITIALIZATION AND PLANNING PHASE

Date	29-09-2026
Team ID	SWTID-2026-6684
Project Title	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The proposal report aims to revolutionize comic story creation and visual storytelling using generative AI and Google Gemini models, boosting creative efficiency, visual narration consistency, and accessibility. It addresses traditional storyboarding and illustration bottlenecks, promising seamless content production, reduced turnaround times, and immersive user experiences. Key features include Gemini-driven multimodal script generation, automated panel breakdown, character design framing, and real-time interactive story branching.

PROJECT OVERVIEW	
Objective	The primary objective is to transform the comic creation workflow by leveraging advanced Google Gemini multimodal models, enabling automated scriptwriting, scene composition, visual prompt creation, and cohesive comic panel generation.
Scope	The project comprehensively assists creators, writers, and artists by structuring story ideas, auto-generating stylized panel descriptions and dialogue, maintaining character visual consistency, and providing an end-to-end interactive digital publishing tool.

PROBLEM STATEMENT	
Description	Addressing time constraints, high production costs, and artistic barriers in traditional comic creation, which limit aspiring storytellers and impede rapid content iteration.
Impact	Solving these bottlenecks will democratize visual storytelling, significantly reduce comic production cycles, lower creative overhead, and enhance user engagement across digital platforms.

PROPOSED SOLUTION	
Approach	Employing Google Gemini models (Gemini 1.5 Pro / Flash) to generate detailed comic scripts, dynamic layout prompts, speech bubbles, and character context embeddings within a web-based studio environment.
Key Features	• Implementation of Gemini-driven automated comic script and panel generation. • Real-time multimodal prompt engineering for consistent artwork and character continuity. • Dynamic dialogue generation, customizable story branching, and visual layout export options.

Resource Requirements

Resource Type	Description	Specification/Allocation
HARDWARE		
Computing Resources	CPU/GPU specifications, memory, cores	NVIDIA RTX 4090 / Cloud GPU Instance (16GB VRAM)
Memory	RAM specifications	16 GB DDR5 RAM
Storage	Disk space for models, assets, and cache	512 GB NVMe SSD
SOFTWARE		
Frameworks	Application framework	Python, FastAPI / Streamlit / React
Libraries	AI / ML & utility libraries	google-generativeai, pillow, pandas, numpy, diffusers
Development Environment	IDE / API access	VS Code, Jupyter Notebook, Google AI Studio API
DATA		
Data	Source, size, format	Custom Comic Script Templates, Multi-genre Story Prompts, Structured JSON Datasets